

Sales Data Analysis Dashboard Report

1. Introduction

This project focuses on analyzing sample sales data from the years 2003 to 2005 using Power BI. The objective is to transform raw sales data into meaningful insights through interactive visualizations. The dashboard enables users to efficiently explore sales performance, order trends, and regional distribution.

2. Objectives

- To analyze overall sales performance across multiple years.
- To identify trends in revenue, orders, and quantity sold.
- To understand country-wise contribution to total sales.
- To build an interactive dashboard for better decision-making.

3. Data Overview

The dataset contains key business information such as:

- Sales revenue
- Order numbers
- Quantity ordered
- Year of transaction
- Country and product categories

Data cleaning was performed to remove duplicates, handle missing values, and ensure consistency.

4. Tools and Techniques Used

- **Power BI** for data visualization and dashboard creation
- **Data Transformation (Power Query)** for cleaning and structuring data

- **DAX (Data Analysis Expressions)** for creating calculated measures such as:
 - Total Sales
 - Total Orders
 - Total Quantity

5. Dashboard Features

The dashboard includes the following components:

5.1 KPI Cards

- Total Sales
- Total Orders
- Total Quantity

These provide a quick overview of business performance.

5.2 Slicers (Filters)

- Year slicer (2003–2005)
- Country slicer

These allow users to filter and analyze data interactively.

5.3 Visualizations

- **Line Chart:** Displays sales trend over time
- **Bar Chart:** Compares sales across categories or countries
- **Pie Chart:** Shows distribution of sales by country

6. Key Insights

- Sales peaked in **2004**, indicating the highest business performance during the period.
- **2003 shows relatively lower sales**, suggesting an early growth stage.
- Sales distribution is concentrated in **a few key countries**, highlighting market dependency.

- There is a variation between the number of orders and total sales, indicating differences in average order value.
- Certain product categories contribute significantly more to revenue than others.

7. Conclusion

The dashboard successfully transforms raw data into actionable insights. It enables users to monitor sales trends, compare regional performance, and identify key drivers of revenue. The interactive design improves usability and supports data-driven decision-making.

8. Future Improvements

- Add customer-level analysis (Top customers)
- Include monthly trends for deeper time analysis
- Enhance visuals with advanced formatting and drill-through features

9. Summary

This project demonstrates the effective use of Power BI for data analysis and visualization. It highlights the importance of data-driven insights in understanding business performance and making informed decisions.

